

{ Practice Question 1

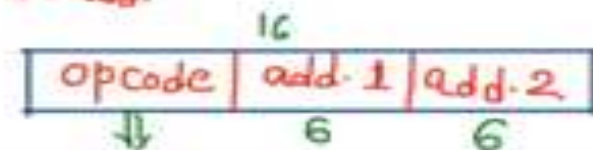
- Consider a digital computer which supports only 2-address instructions each with 16-bits. If address length is 6-bits then maximum and minimum how many instructions the system can support?

16

1

+

2- add.



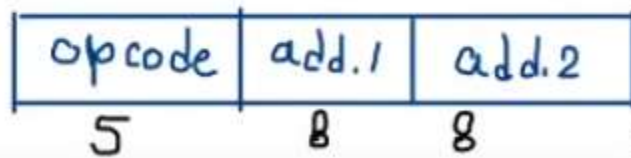
4-bits

→ max opcode = $2^4 = 16$

Practice Question 2

- Consider a digital computer which supports 32 2-address instructions. If address length is 8-bits then the length of the instruction is 21 bits?

$$\text{No. of instructions} = 32 \Rightarrow \text{opcode bits} = \log_2 32 = 5\text{-bits}$$



$$\Rightarrow 5 + 8 + 8 = 21\text{-bits}$$

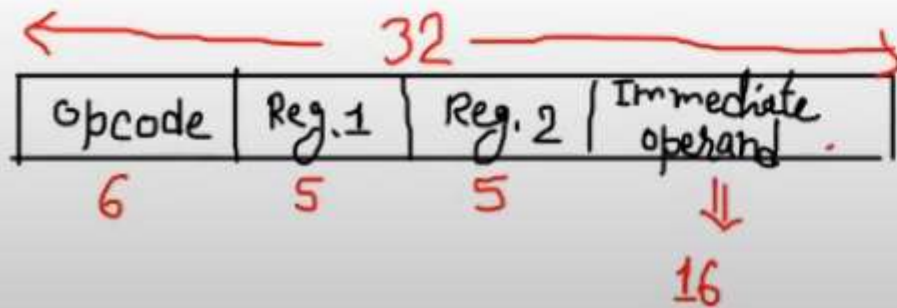
Practice Question 3

{ GATE-2016 }

- A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is 16-bit?

No. of instructions = 40 $\Rightarrow$ Opcode - bits = 6-bits

No. of registers = 24 $\Rightarrow$ Reg. Reference = 5-bits

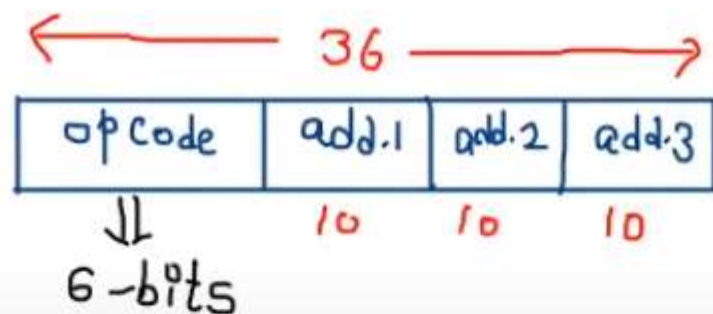


Question 1

Consider a digital computer which supports only 3-address instructions. If size of instructions is 36-bits and the size of each address is 10-bits, then minimum and maximum number of instructions the computer can support?

1

64



$$\text{max op codes} = 2^6 = \underline{\underline{64}}$$

+

Question 2

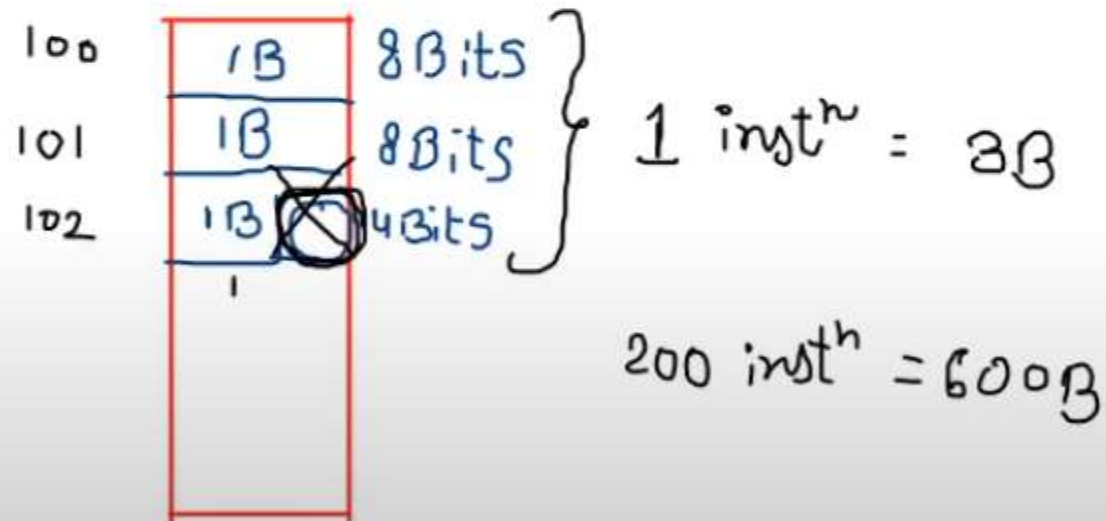
Consider a digital computer which supports 16 2-address instructions. If size of each address is 8-bits, then the size of instruction is _____ bits?

$$\text{No. of instructions} = 16 \Rightarrow \text{opcode-bits} = 4 \left(\log_2 16 \right)$$



Question 3

In previous question if the instructions are stored in memory in byte-aligned fashion then space required in memory to store a program text of 200 instructions is 600 bytes?



+

Practice Question 5

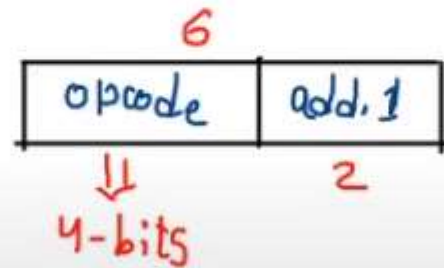
(Multiple Instⁿ Support)
2 questⁿ in 3 year.

- Consider a system which supports 2-address instructions and 1-address instructions both. The system has 6-bit instructions and 2-bits addresses. If there are three 2-address instruction in the system then maximum and minimum how many 1-address instructions the system can support?

2-add.



1-add.



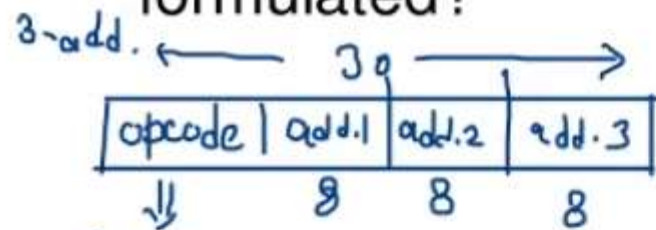
$$\begin{aligned} \text{max opcodes} &= 2^2 = 4 (00, 01, 10, 11) \\ \text{used opcode} &= 3 (\text{assume: } 00, 01, 10) \\ \hline \text{unused opcode} &= 1 (11) \end{aligned}$$

1100
1101
1110
1111

} max for 1-address instⁿs

Practice Question 6

- Consider a system which supports 3-address and 2-address instructions both. It has 30-bit instructions with 8-bit addresses. If there are 'x' 3-address instructions then maximum how many 2-address instructions can be formulated?



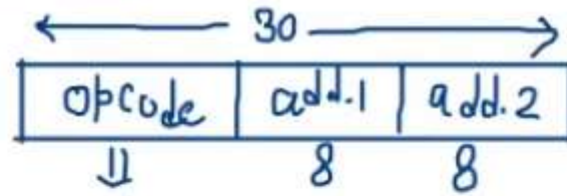
6-bits

max opcodes = $2^6 = 64$

used opcodes = x

unused opcodes = $(64 - x)$

2-add.



14-bits

6 8

$(64 - x) * 2^8$

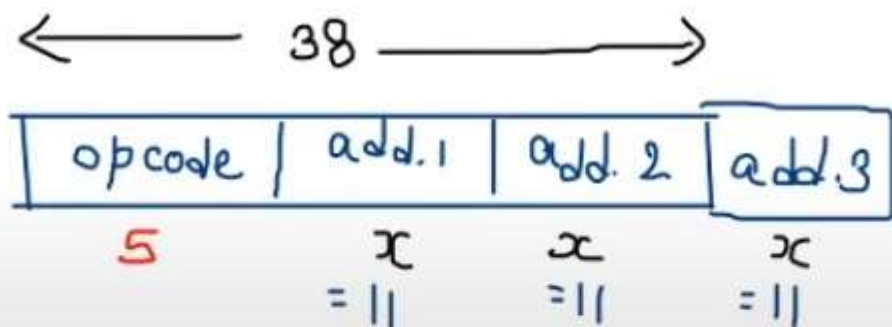
max 2-add.
instructions
possible

Question 4

+

Consider a digital computer which supports 32³ 2-address instructions. Size of each instruction is 38-bits. The system supports a word addressable memory, with word size of 32-bits, then the maximum allowable size of the memory is 8 Kbytes?

No. of opcode = 32 $\Rightarrow$ opcode bits = 5-bits



$$5 + x + x + x = 38 \text{ bits}$$

$$x = 11 \text{ bits}$$

$$2^{11} \times 32 \text{ bits}$$

$$2^{11} \times 4 \text{ Bytes}$$

$$8 \text{ KB}$$

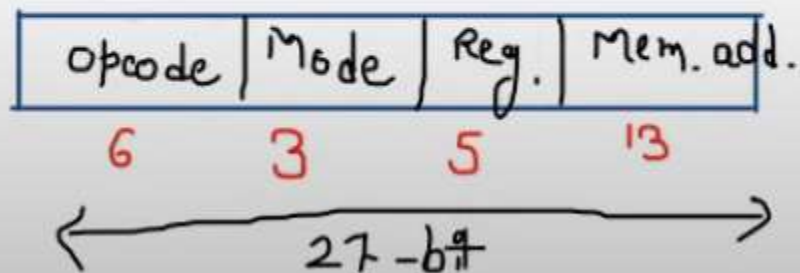
Question 5

Consider a digital computer which supports maximum of 32KB word addressable memory with word size 32-bits. The computer supports 40 distinct instructions each of which has 4 fields: 4B

$$\text{No. of instructions} = 40 \Rightarrow \text{opcode} = \lceil \log 40 \rceil = 6 \text{ bits}$$

1. Opcode
2. Mode field to specify one of 7 addressing modes
3. Register field to specify one of 24 processor registers
4. Memory address field

The size of instruction is _____ bits?



$$\text{No. of words in mem.} = \frac{32\text{KB}}{4\text{B}}$$

$$= 8\text{k words}$$

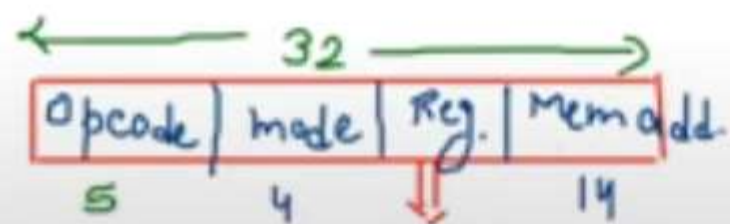
$$\text{add.} = 13 \leftarrow (= 2^{13})$$

ISA = Instruction Set Architecture Question 6 + No. of instructions = 18 $\Rightarrow$ opcode = 5

Consider a digital computer which supports 32-bits instructions has instruction set of size 18, with each instruction has 4 fields:

1. Opcode
2. Mode field to specify one of 10 addressing modes $\Rightarrow$ mode-bits = 4
3. Register field
4. Memory address field of 14-bits

The maximum number of general purpose registers possible in the CPU is?

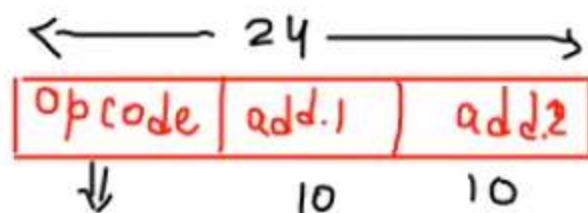


$$32 - 23 = 9 \Rightarrow \max \text{ Reg.} = 2^9 = 512$$

Question 7 MSQ

Consider a system which supports 2-address instructions. If system has 24-bits instructions and 10-bits addresses then which of the following is/are possible number of instructions supported by system?

- a) 0
- ☒ b) 1
- ☒ c) 8
- ☒ d) 16



4-bits $\Rightarrow$ max opcodes = $2^4 = 16$

Range :- 1 To 16

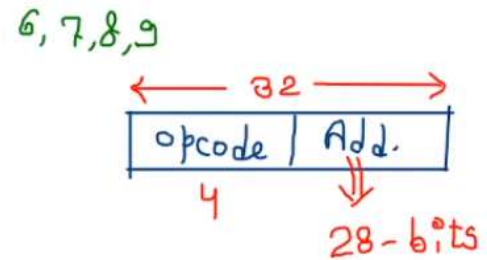
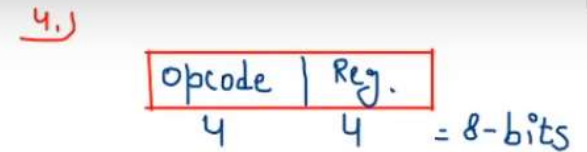
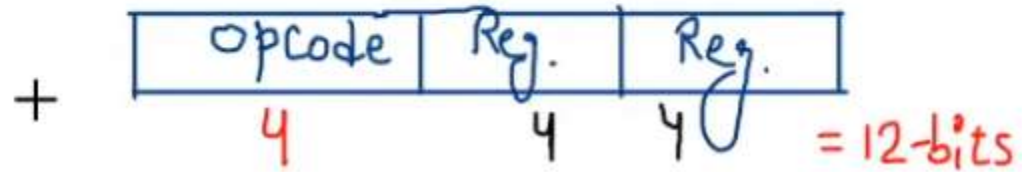
Reg. Ref. = 4-bits

Question 8

Total inst^{ns} = 9 $\Rightarrow$ fixed opcode = 4-bit

Consider a hypothetical processor with largest instruction length being 32-bits and total 16 registers R0-R15. Processor supports following instructions:

- | | | |
|----------|------------|-----------------------------|
| 1. ADD | R_i, R_j | 1, 2, 3, 5 |
| 2. SUB | R_i, R_j | |
| 3. AND | R_i, R_j | |
| 4. NOT | R_i | |
| 5. MOV | R_i, R_j | |
| 6. LOAD | Address | // Loads with register R0 |
| 7. Store | Address | // Stores the content of R1 |
| 8. JUMP | Address | |
| 9. JZ | Address | |



The maximum number of address lines on this processor is 28 ?

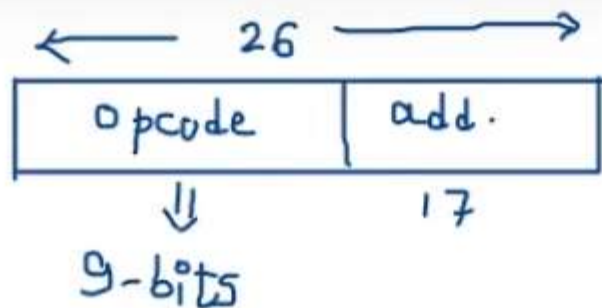
Question 9

$$2^{17} \Rightarrow \text{mem add.} = 17\text{-bits}$$

A system supports 1-address and 0-address instructions with the instruction size of 26-bits. The system supports a byte addressable 128KBytes memory. If there are 500 1-address instructions then maximum how many 0-address instructions can be supported in the system?

+

1-add.

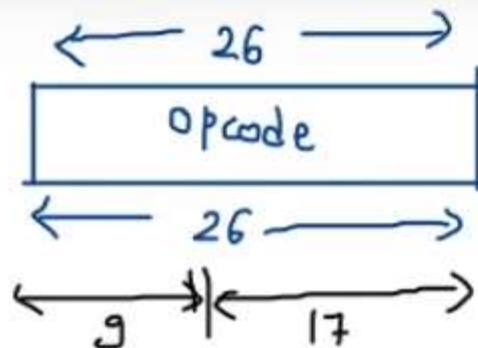


$\max \text{ opcode} = 2^9 = 512$

used opcodes = 500

unused opcodes = 12

0-add.



unused

$\max = 12 \times 2^{17}$

+

Question 10

What is the range of opcodes for 1-address and 0-address instructions in above question?

1-add. instⁿ $\Rightarrow$ 500 to 500

0-add. instⁿ $\Rightarrow$ 1 to $12 * 2^{17}$

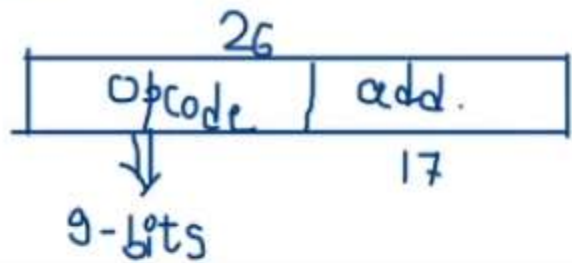
Ques) $Inst^n = 26\text{-bits}$

$add. = 17\text{-bits}$

1-add., 0-add. instructions supported.

Range of 1-add., 0-add. instructions?

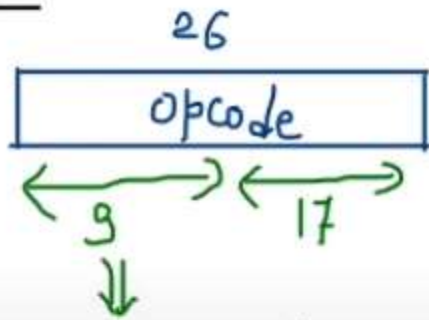
1-add.:-



$$\text{max opcode} = 2^9 = 512$$

used opcodes $\Rightarrow 1$ to 511

0-add.



$$511 \times 2^{17} = \text{max}$$

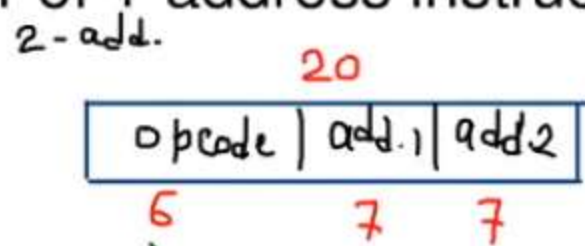
1-add. $\Rightarrow 1$ to 511

0-add. $\Rightarrow 1$ to 511×2^{17} +

Question 11 MSQ

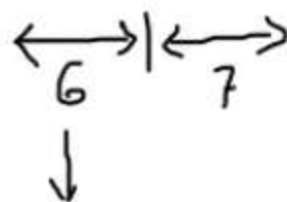
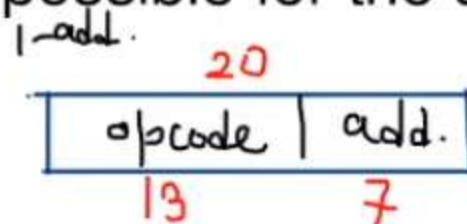
Consider a system which supports 2-address and 1-address instructions both. System has 20-bits instructions and 7-bits addresses then which of the following is/are the number of 1-address instructions possible for the system?

- a) 0
- b) 8065
- c) 8192
- d) 128



$\downarrow$
 $\max = 2^6 = 64$

$\text{used} = 1$
 $\hline \text{unused} = 63$



$63 \times 2^7 = 8064 = \max$

Range for 1-add instⁿ
1 to 8064

Question 14

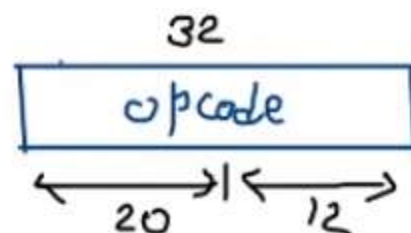
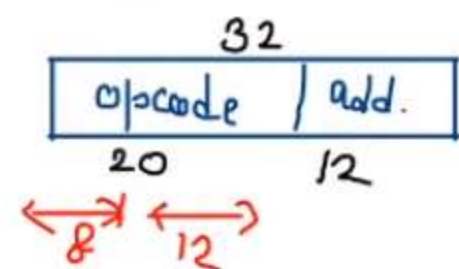
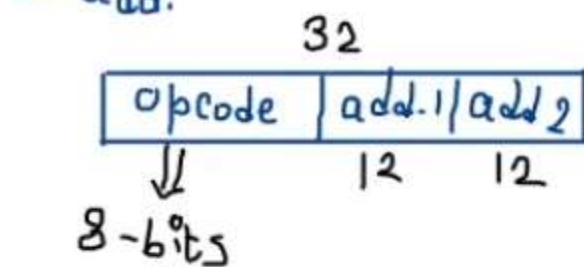
1. 8192
2. $192 * 2^{12}$

+

Consider a system with 32-bit instructions and 12-bit addresses. If there are 254 2-address instructions then maximum how many 1-address instructions can be formulated in the system? Suppose the same system has 8000 1-address instructions then maximum how many 0-address instructions can be formulated?

2-add.

1-add.



$$\begin{aligned} \text{max} &= 2^8 = 256 \\ \text{used} &= 254 \\ \hline \text{unused} &= 2 \end{aligned}$$

unused

$$2 * 2^{12} = 2^{13} = 8192$$

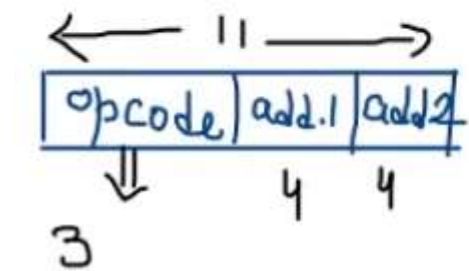
$$\begin{aligned} &8192 \\ &- 8000 \\ \hline \text{unused} &= 192 \end{aligned}$$

unused

$$192 * 2^{12}$$

In an 11-bit computer instruction format, the size of address field is 4-bits. The computer uses expanding OP code technique and has 5 two-address instructions and 32 one-address instructions. The number of zero-address instructions it can support is 256

2-add.

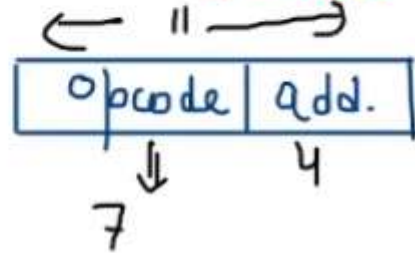


$$\text{max opcode} = 2^3 = 8$$

$$\text{used opcode} = 5$$

$$\text{unused opcode} = 3$$

1-add.

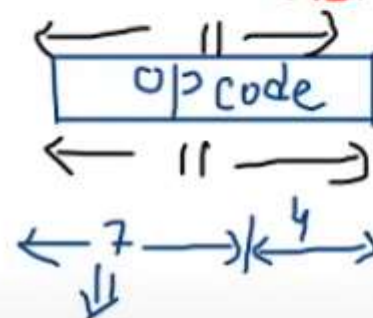


$$\text{max} = 3 * 2^4 = 48$$

$$\text{used} = 32$$

$$\text{unused} = 16$$

0-add.



$$\text{max} = 16 * 2^4$$

$$= 256$$

CPU can process
32-bit data

1 word = 32-bits

A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is _____

Instruction size = 32-bits

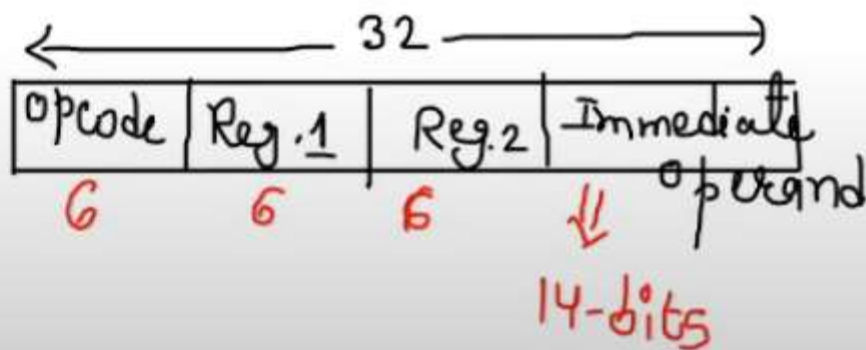
No. of regs = 64 $\Rightarrow$ Reg. Reference = $\log 64 = 6$ -bits

$$\begin{aligned} ||| &= \text{max} \\ &= 7 \\ ||| &= 15 \end{aligned}$$

No. of instⁿ = 45

opcode-bits

$$\lceil \log_2 45 \rceil = 6\text{-bits}$$



$$\text{max value} = 2^{14} - 1 = 16383$$